

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Physics — Class IX

**Syllabus: Ch 1 — Physical Quantities & Measurement | Ch 2 — Kinematics | Ch 3 — Dynamics-I
| Ch 4 — Dynamics-II | Ch 5 — Pressure & Deformation | Ch 6 — Work & Energy (till 6.4) | Ch
9 — Nature of Science**

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

1. Write your name, roll number and class clearly on the answer sheet before starting.
2. Attempt all questions from Section A (MCQs). In Sections B and C, attempt every question or its alternative.
3. Use $g = 9.8 \text{ m/s}^2$ (or $g = 10 \text{ m/s}^2$ where stated) in numerical problems unless otherwise specified.
4. Mobile phones and electronic devices are not permitted in the examination hall.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

#	Question	A	B	C	D	Answer
1	The SI unit of force is the newton. One newton equals:	Ⓐ kg·m/s	Ⓑ kg·m/s ²	Ⓒ kg·m ² /s ²	Ⓓ kg/m·s ²	
2	Least count of a vernier caliper is:	Ⓐ 0.001 cm	Ⓑ 0.01 cm	Ⓒ 0.1 cm	Ⓓ 1 cm	
3	A ball thrown vertically upward decelerates at $g = 9.8 \text{ m/s}^2$. This is an example of:	Ⓐ Uniform motion	Ⓑ Zero acceleration	Ⓒ Negative acceleration	Ⓓ Positive acceleration	
4	Newton's first law of motion defines the concept of:	Ⓐ Momentum	Ⓑ Inertia	Ⓒ Friction	Ⓓ Weight	
5	The moment of a force about a pivot equals $F \times d$. The SI unit of torque is:	Ⓐ N/m	Ⓑ N·m	Ⓒ N·m ²	Ⓓ kg·m/s	
6	Hooke's law states that extension of a spring is directly proportional to the applied force, provided the force does not exceed the:	Ⓐ Breaking point	Ⓑ Elastic limit	Ⓒ Yield strength	Ⓓ Plastic region	
7	Pascal's principle states that pressure applied to an enclosed fluid is transmitted:	Ⓐ Only upward	Ⓑ Only downward	Ⓒ Equally in all directions	Ⓓ Only horizontally	
8	The kinetic energy of a body of mass m moving at speed v is:	Ⓐ mv	Ⓑ mv^2	Ⓒ $\frac{1}{2}mv^2$	Ⓓ mgh	
9	A 5 kg object is lifted to a height of 4 m. Its gravitational potential energy ($g = 10 \text{ m/s}^2$) is:	Ⓐ 20 J	Ⓑ 100 J	Ⓒ 200 J	Ⓓ 400 J	
10	Power is defined as the rate of doing work. Its SI unit is:	Ⓐ Joule	Ⓑ Newton	Ⓒ Watt	Ⓓ Pascal	
11	A scientific theory is best described as:	Ⓐ A guess about nature	Ⓑ A tested explanation of natural phenomena	Ⓒ An unverified hypothesis	Ⓓ A laboratory observation	
12	Which of the following is a vector quantity?	Ⓐ Speed	Ⓑ Mass	Ⓒ Temperature	Ⓓ Velocity	

SECTION B — Short Questions ($11 \times 3 = 33$ Marks)

Attempt all 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Differentiate between base quantities and derived quantities. Give two examples of each with their SI units.	3	OR	Explain the terms scalars and vectors. Give two examples of each. How are vectors represented graphically?	3

2(ii)	Define distance and displacement. A student walks 3 km North then 4 km East. Find the total distance and magnitude of displacement.	3	OR	Define speed and velocity. Distinguish between uniform and variable velocity with examples. What does the gradient of a distance-time graph represent?	3
2(iii)	State Newton's second law of motion. A net force of 60 N acts on a 12 kg object. Calculate the acceleration produced. Give the SI unit of force derived from this law.	3	OR	Define momentum and impulse. Show that impulse equals the change in momentum. Give the SI unit of momentum.	3
2(iv)	State Newton's third law of motion. Give two real-life examples and identify the action-reaction pairs in each example.	3	OR	State the law of conservation of momentum. A gun of mass 3 kg fires a bullet of mass 0.03 kg at 300 m/s. Calculate the recoil velocity of the gun.	3
2(v)	Define torque (moment of force). State the principle of moments. A spanner applies a force of 40 N at a perpendicular distance of 0.25 m from a bolt. Calculate the torque.	3	OR	Define equilibrium and state its two conditions. Distinguish between static equilibrium and dynamic equilibrium with one example each.	3
2(vi)	Differentiate between stable, unstable and neutral equilibrium. How does the position of the centre of gravity affect the stability of an object?	3	OR	Explain friction and its types. Give two advantages and two disadvantages of friction in everyday life.	3
2(vii)	State Hooke's law. Define spring constant. A spring extends 5 cm when a force of 20 N is applied. Calculate the spring constant.	3	OR	Define pressure. Explain how pressure varies when (a) force is kept constant but area changes, (b) area is kept constant but force changes. Give a practical example for each.	3
2(viii)	Define atmospheric pressure. Explain how it is measured by a mercury barometer. What is the standard atmospheric pressure in Pa?	3	OR	Derive the formula for pressure at depth h in a liquid: $P = \rho gh$. How does liquid pressure change with depth?	3
2(ix)	Define work done. A force of 150 N pushes a box through 5 m on a frictionless floor. Calculate the work done. Under what condition is work done zero even when force is applied?	3	OR	State the law of conservation of energy. A ball of mass 2 kg is dropped from 10 m height. Find (a) potential energy at the top, and (b) kinetic energy just before hitting the ground. ($g = 10 \text{ m/s}^2$)	3
2(x)	Define power and efficiency. A machine does 4800 J of useful work using 6000 J of input energy in 20 s. Calculate (a) efficiency, and (b) power output of the machine.	3	OR	Differentiate between renewable and non-renewable energy resources. Give two examples of each. Why are renewable sources preferred for the future?	3
2(xi)	Define hypothesis, theory and law as used in scientific research. Give one well-known example of each from physics.	3	OR	Differentiate between science, technology and engineering with suitable examples from daily life. How are the three fields interconnected?	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Define acceleration and derive the equations of motion for uniformly accelerated motion graphically using a speed-time graph. State two key points read from a speed-time graph.	5	OR	Explain the types of motion: translatory (rectilinear, curvilinear, random), rotatory and vibratory. Differentiate between distance and displacement, and between speed and velocity. Give one daily-life example of each type of motion.	5
3(ii)	State Newton's three laws of motion. For each law: write a mathematical statement where applicable, and give one practical example. Also state the limitations of Newton's laws.	5	OR	Define momentum. Derive the relationship between force and rate of change of momentum. A car of mass 1200 kg moving at 8 m/s brakes to a stop in 5 s. Calculate the average braking force using $F = \Delta p / \Delta t$.	5
3(iii)	State Pascal's principle. Explain the working principle of a hydraulic lift. A hydraulic lift has a narrow piston area of 0.002 m^2 and a wider piston area of 0.9 m^2 . A car weighing 17,640 N is to be lifted. Find the force needed at the narrow piston.	5	OR	Define elasticity and Hooke's law. Sketch and explain the force-extension graph for an elastic solid, labelling the limit of proportionality, elastic limit and breaking point. State three practical applications of Hooke's law.	5
3(iv)	Define kinetic energy and gravitational potential energy. Derive expressions for both from first principles. State the law of conservation of energy and show using a falling ball example that total mechanical energy remains constant at all points of fall.	5	OR	Define power and efficiency. A petrol engine takes in 400 J of fuel energy and delivers 120 J of useful work. (a) Calculate its efficiency as a percentage. (b) If the engine completes this cycle in 0.5 s, find the useful power output. (c) Explain why a machine can never be	5

				100% efficient.	
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Invigilator’s Signature: _____

Candidate’s Signature: _____

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— *End of Question Paper* —